MINOR PROJECT REPORT

Predictive Classification Using Random Forest Ensembles

Customer Churn Prediction Using the Telco Customer Churn Dataset

Algorithm: Random Forest Classifier

Language: Python **Platform:** Google Colab / Jupyter Notebook

Prepared from the uploaded dataset and aligned with the project guidelines for EDA, preprocessing, evaluation metrics, visualizations, and conclusion.

Abstract

This project builds a binary classification model to predict customer churn from telecommunications customer data. The workflow includes dataset inspection, exploratory data analysis, missing-value handling, categorical encoding, a stratified 80:20 train-test split, Random Forest training, and evaluation using accuracy, precision, recall, F1-score, ROC-AUC, a confusion matrix, and feature importance. The report records measured performance from the uploaded CSV. The model is an educational predictive analytics demonstration and should not be treated as a certainty about individual customer behavior.

1. Introduction

Customer churn occurs when a customer stops using a company's service. Predicting churn can help organizations investigate customer needs and improve retention strategies. Machine learning classification learns patterns from historical customer records and estimates whether a record belongs to the churn or non-churn class. Random Forest is an ensemble algorithm that combines multiple decision trees to produce a final prediction.

2. Problem Statement

The objective is to classify customer records as churn (Yes) or no churn (No) using available demographic, service, contract, tenure, and billing information. The target column is **Churn**; No is encoded as 0 and Yes as 1.

3. Objectives

• Inspect and understand the dataset.

• Perform EDA on churn and selected features.

• Clean missing or invalid values and prepare numerical and categorical features.

• Train a Random Forest classifier with an 80:20 stratified split.

• Evaluate using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix.

• Identify features with high model importance scores.

• Discuss limitations and potential improvements.

4. Dataset Description

The uploaded CSV contains **7,043 rows** and **21 columns** before preprocessing. The target column is **Churn**. The dataset includes customer profile, subscribed services, contract/payment details, tenure, and billing information. An ID column, if present, is excluded from model features because it identifies records rather than describing customer behavior.

Item	Description
Rows	7,043
Columns	21
Target	Churn
Target encoding	No = 0; Yes = 1
Split	80% training / 20% testing

5. Methodology

Workflow: dataset loading → inspection → EDA → cleaning → feature/target separation → train-test split → preprocessing pipeline → Random Forest training → prediction → evaluation → interpretation. The preprocessing pipeline is fitted on training data to reduce leakage from the test set.

5.1 Data cleaning and preprocessing

TotalCharges, when present, is converted to numeric values. Blank or invalid entries become missing values. Numerical missing values are imputed with the median; categorical missing values are imputed with the most frequent category.

Customer ID is excluded. Categorical features are transformed using One-Hot Encoding, and the target is mapped to 0/1.

5.2 Random Forest settings

Parameter	Setting	Reason
n_estimators	200	Combines 200 decision trees.
max_depth	None	No fixed maximum depth is imposed.
class_weight	balanced	Adjusts weights based on class frequencies.
random_state	42	Supports reproducibility.
n_jobs	-1	Uses available CPU cores.

Feature scaling is not applied because Random Forest is tree-based and does not rely on distance calculations. The data is split into 80% training and 20% testing using stratification and random_state=42.

6. Exploratory Data Analysis

The target distribution is examined to identify class imbalance. Histograms summarize selected numerical features, and a correlation heatmap displays linear relationships among numerical columns and the encoded target. Correlation does not capture all relationships involving categorical variables.

6.1 Churn distribution

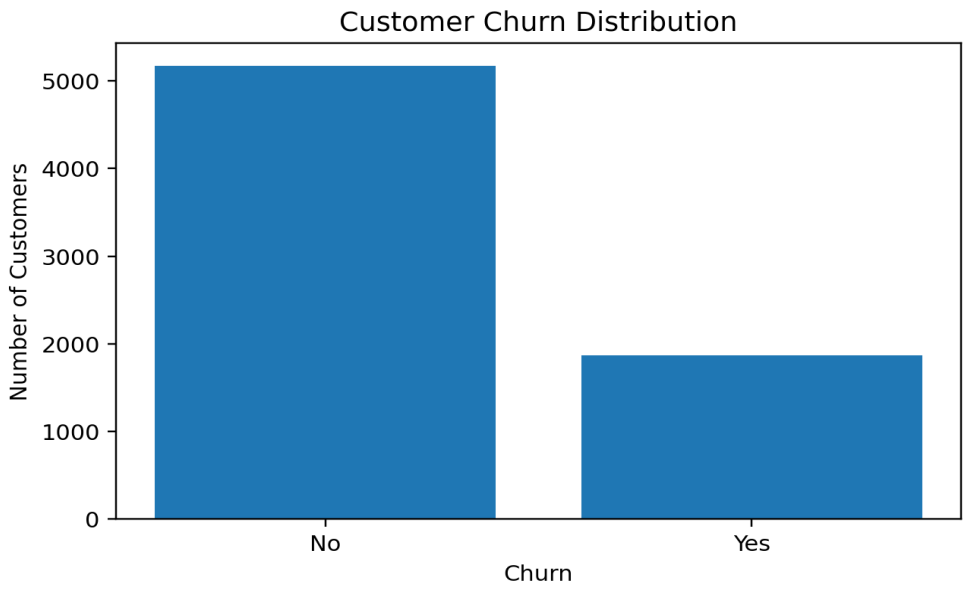


Figure 1. Distribution of churn target classes.

6.2 Numerical feature distributions

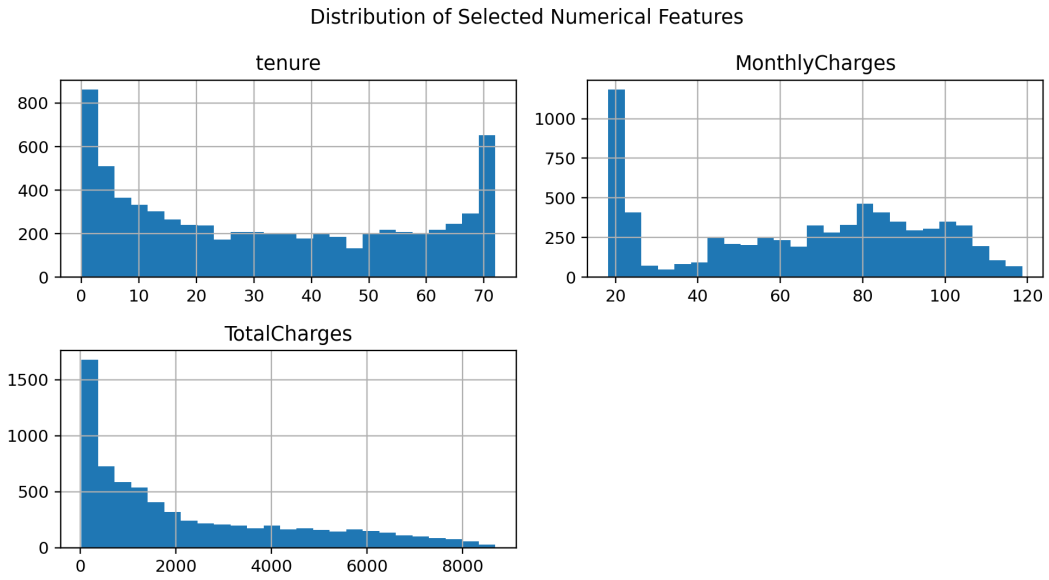


Figure 2. Histograms of selected numerical features.

6.3 Correlation heatmap

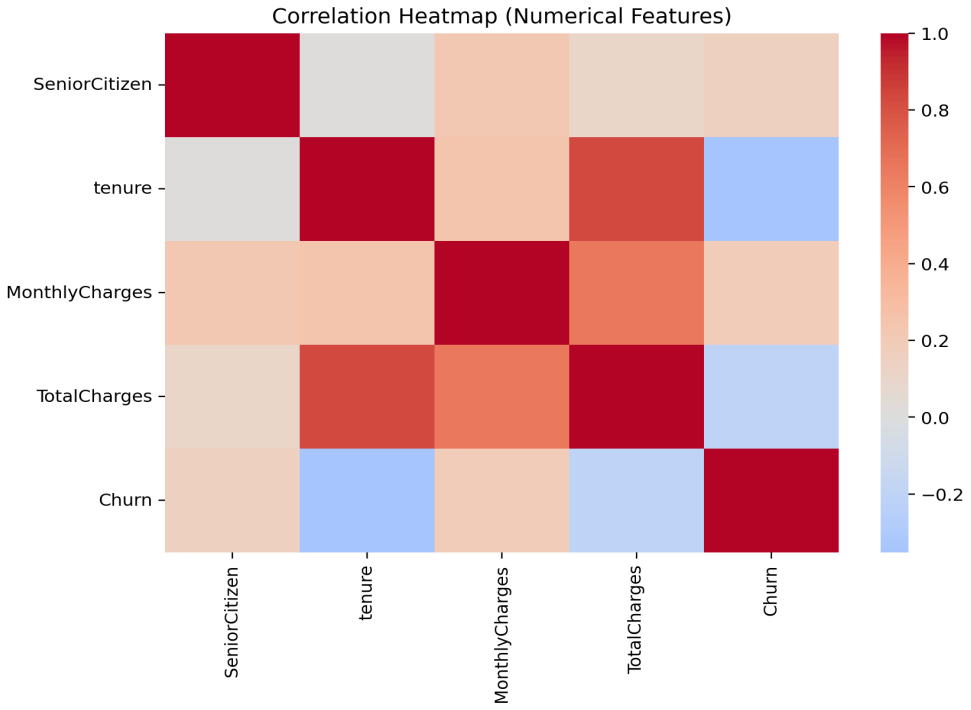


Figure 3. Correlations among numerical columns and encoded churn.

7. Model Evaluation

Churn (Yes, encoded as 1) is treated as the positive class. Precision measures how many predicted churners truly churned. Recall measures how many actual churners were identified. F1-score balances precision and recall. ROC-AUC evaluates ranking/discrimination across classification thresholds.

Metric	Measured result	Interpretation
Accuracy	0.7842 (78.42%)	Fraction of all test records classified correctly.
Precision (Churn)	0.6224	Share of predicted churners that actually churned.
Recall (Churn)	0.4759	Share of actual churn cases detected.
F1-score (Churn)	0.5394	Harmonic mean of churn precision and recall.

ROC-AUC	0.8226	Measures discrimination across thresholds.

7.1 Confusion matrix

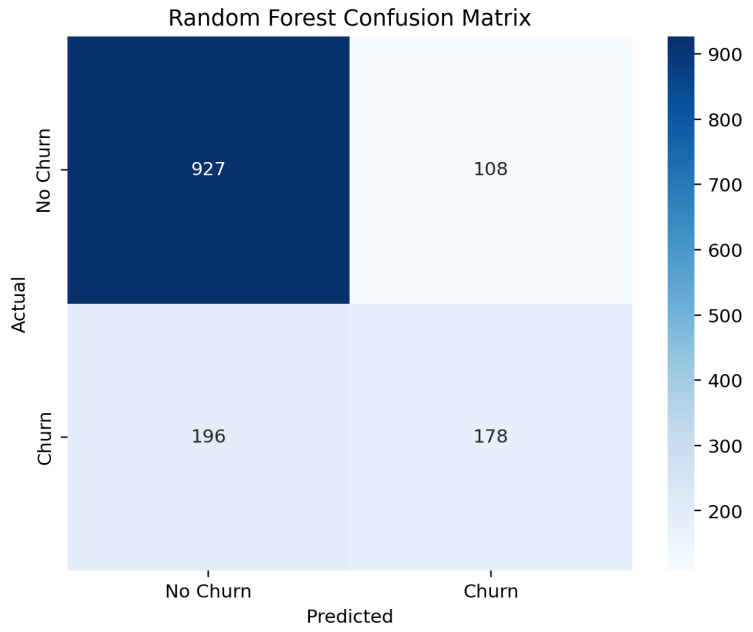


Figure 4. Confusion matrix. TN = 927, FP = 108, FN = 196, TP = 178. A false negative is a churner incorrectly predicted as no churn; a false positive is a non-churner incorrectly predicted as churn.

7.2 ROC curve

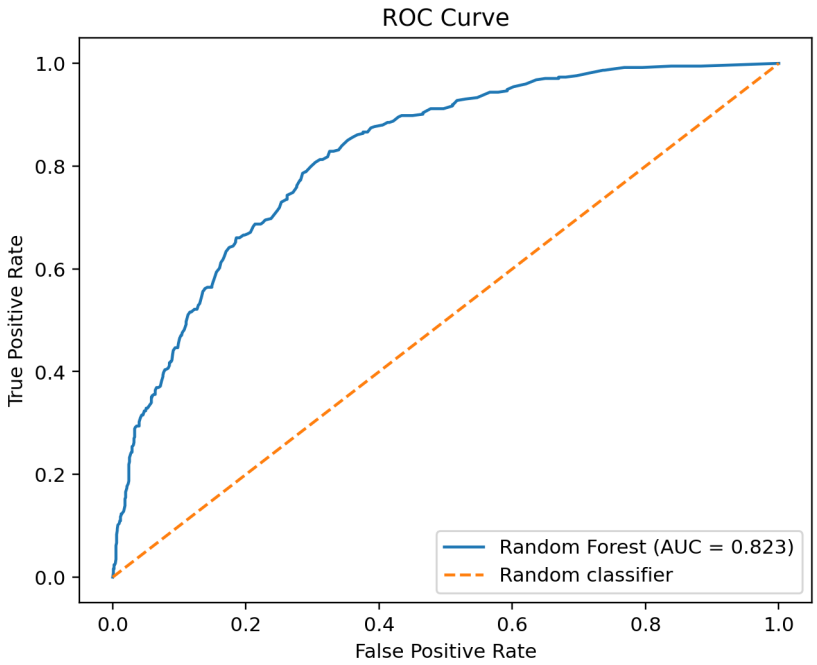


Figure 5. ROC curve with measured AUC.

7.3 Feature importance

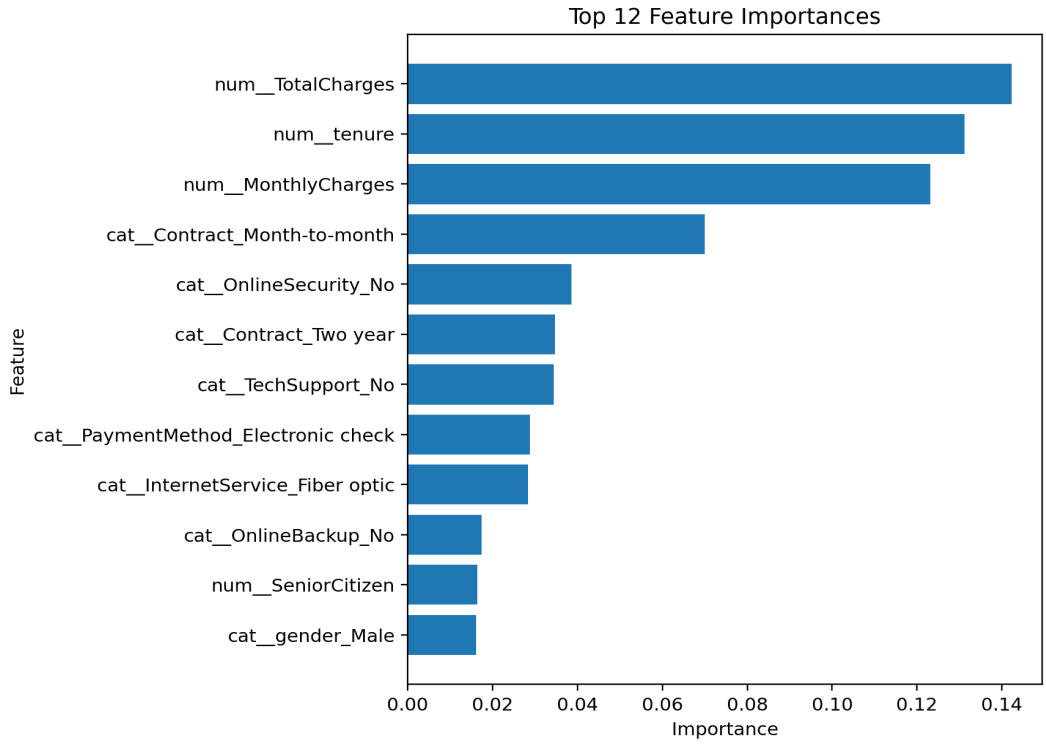


Figure 6. Top 12 features by Random Forest importance. These scores describe how the model used features and do not establish causation.

8. Results and Discussion

On the held-out test set, the model achieved accuracy 0.7842, churn precision 0.6224, churn recall 0.4759, churn F1-score 0.5394, and ROC-AUC 0.8226. These metrics should be considered together. If the business objective is to find as many potential churners as possible, recall is important, although increasing recall can also increase false positives and unnecessary retention outreach.

The feature-importance chart ranks features according to their contribution to the fitted model's tree decisions. Importance can be affected by correlated features and should not be interpreted as proof that a feature causes churn.

9. Limitations

- Historical patterns may not represent future customer behavior.
- One train-test split provides only one performance estimate.
- Class imbalance can make accuracy alone misleading.
- Feature importance is not a causal explanation.
- Predictions should support, not replace, business judgment.

10. Future Enhancements

Possible enhancements include tuning max_depth, min_samples_split, min_samples_leaf, and class_weight; comparing Random Forest with Logistic Regression or Gradient Boosting; selecting a decision threshold based on business costs; performing cross-validation; and validating the model on newer customer data.

11. Conclusion

This project demonstrates an end-to-end customer churn classification workflow using a Random Forest ensemble. It includes EDA, data cleaning, preprocessing, an 80:20 split, model training, and evaluation with the required metrics and visualizations. The measured test results provide a baseline for future experimentation. Further work can focus on improving churn detection, comparing models, tuning hyperparameters, and validating on future customer records. The model should be treated as a decision-support tool rather than a guarantee about individual customer actions.

Appendix: Reproducibility

Results were generated from the uploaded CSV using a stratified 80:20 split, random_state=42, 200-tree Random Forest, balanced class weights, median numerical imputation, most-frequent categorical imputation, and One-Hot Encoding. Keep these settings aligned with the notebook when submitting.